

Functions and Function Notation

Name _____

Circuit

Begin with the problem in the top left. Answer the question. To advance in the circuit, find your answer and write “2” in the blank. Solve that problem, find your answer and write “3” in the blank and so on until you have solved every problem and returned to the beginning.

ANSWER: 4

_____ Given $f(x) = x^2 - 3x + 5$, find $f(2)$.

ANSWER: $2a + 5$

_____ Given $g(x) = x^3$, find $g(2)$.

ANSWER: 5

_____ The function f is given by the table below.
Find $f(4)$.

x	1	2	3	4	5
$f(x)$	8	6	7	6	8

ANSWER: 2

_____ Given $f(x) = \begin{cases} 4x - 5, & x \leq 2 \\ x^2 - 5, & x > 2 \end{cases}$, find $f(3)$.

ANSWER: 8 # _____ Given $h(x) = 3x - 1$, write a simplified expression for $h(x + 3)$.	ANSWER: 6 # _____ Using the same table, how many inputs x satisfy $f(x) = 6$?
ANSWER: $3x + 8$ # _____ Given $f(x) = x^2 - 3x + 5$, find $f(0)$.	ANSWER: -5 # _____ Given $f(x) = 2x + 3$, write a simplified expression for $f(a + 1)$.
ANSWER: 9 # _____ Given $f(x) = 2x + 3$, find $f(-4)$.	ANSWER: 3 # _____ Given $f(x) = x^2 - 3x + 5$, find $f(-1)$.